Control Systems (EEE/ECE/INSTR F242)

Tutorial 6 (17 February 2026)

1. Consider a DC servomotor driving a load as shown in Fig. Q1. The torque-speed curve for this motor is given by $T_m = -8\omega_m + 200$, for an input voltage of 100 V.

(i) Find the time taken by the load to reach 75% of its steady state angular speed if input voltage (e_a) instantaneous changes from 0 V to 100 V.

(ii) What will be the effect on the time constant of the system transfer function $(\omega_L(s)/E_a(s))$, if N_4 changes to 125.

Given,

$$\frac{\theta_m(s)}{E_a(s)} = \frac{\frac{K_T}{R_a J_m}}{s(s + \frac{1}{J_m}(D_m + \frac{K_T K_b}{R_a})}$$

where, K_T is motor torque constant, K_b is back emf constant, R_a is armature resistance, J_m and D_m are the equivalent inertia and damping at the motor shaft.

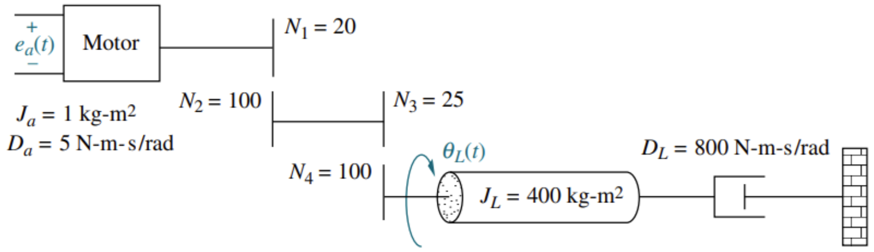

Fig. Q1

2. Given the system G(s) = $C(s)/R(s) = (s+2)/(s+5)$, find the output expression, c(t), for unit impulse and unit step input.

3. A temperature probe (having time constant, 'τ') is transferred from air at 25 °C to an air at 35 °C, and then to water at 70 °C, and back to air at 35 °C. Assume that in each case transfer is instantaneous. Calculate the indicated temperature at the end of each time interval (shown below) and sketch the temperature variation.

Given:

• τ = 30 seconds for dry probe (in air) • τ = 5 seconds in water • τ = 20 seconds for wet probe (in air)	$t < 0$ s, T = 25°C (initial temperature) *0 < t < 7 s,* T = 35 ℃ *(dry probe in air)* *7 < t < 15 s,* T = 70 ℃ *(probe in water)* *15 < t < 30 s,* T = 35℃ *(wet probe in air)*

4. A tank level system is shown in Fig. Q4. If output flow rate of the tank, $Q_0 = \beta\sqrt{h}$, and the cross-section area of the tank is 'A', then find, **(a)** Transfer function, $\Delta H(s)/\Delta Q_i(s)$, where ΔH and ΔQ_i are the deviations from their respective steady state values, **(b)** Find the static gain and time constant of this system around an operating point, $\bar{H}$. **(c)** Calculate the approximate time taken by the liquid level to reach a new steady state, if the operating point $\bar{H}$ is, **(i)** L/2, **(ii)** L/3, where L is the height of the tank.

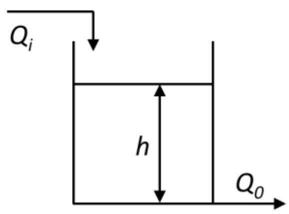

Fig. Q4

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